Aastha Mittal

(602) 706-9102 | aasthami@andrew.cmu.edu | linkedin.com/in/aastha-mittal-cs | github.com/aastha-mittal

Education

Carnegie Mellon University — Pittsburgh, PA
B.S. Computer Science, Concentrations: Systems & Machine Learning/AI — *Expected: May 2028*
Relevant Coursework: Machine Learning (10-601), Generative AI (10-423), Parallel & Sequential Data Structures & Algorithms, Computer Systems, Probability & Statistics

Work Experience

INSIGHT Program — Software Engineering Track — Aug 2026
Jane Street — *New York, NY*
- Selected for Jane Street's competitive INSIGHT program (SWE track), working with full-time developers on software design and trading system fundamentals.

Software Engineer Intern — May 2026 – Aug 2026
IBM Research (Agentic AI Security) — *Durham, NC*
- Architected an LLM-powered policy orchestration system in Go that translated natural-language infrastructure requirements into Terraform configurations for Vault and IAM, reducing developer access configuration time by 50% and eliminating manual policy authoring across enterprise cloud environments.
- Contributing to Smith (github.com/IBM/smith), an open-source skill for AI code agents automating OPA policy creation and LLM-based test generation, helping resolve open issues as part of IBM's agentic AI security work. Monitored infrastructure SLOs with Datadog, triaging production incidents and contributing to post-incident reviews.

Software Engineer Intern — Jan 2026 – May 2026
Time2Cash (Legal Tech Startup, $5M Seed Round) — *Remote*
- Designed and implemented distributed backend services in Python using PostgreSQL, Redis, and Kafka that processed 50K+ legal records daily at 99.2% uptime, applying schema design, indexing, and query tuning to maintain read performance as data volume scaled.
- Built an LLM-powered document analysis pipeline (OpenAI API, LangChain, FAISS) with statistical anomaly detection that reduced manual review time by 65% and surfaced over $2M in revenue leakage across 30+ enterprise portfolios. Instrumented precision and MRR metrics to track model output quality over time.

Teaching Assistant — 15-121: Introduction to Data Structures — Aug 2025 – Current
Carnegie Mellon University — *Pittsburgh, PA*
- Taught data structures, algorithms, and OOP design in Java to 25+ students through weekly lab sections and one-on-one debugging sessions, focusing on correctness, runtime analysis, and engineering fundamentals.

Software Engineer Intern — May 2025 – Aug 2025
Nationwide Insurance — *Columbus, OH*
- Developed a large-scale full-stack application in Java, Spring Boot, Angular, and TypeScript serving over one million customers, maintaining 90%+ test coverage with JUnit and Mockito and participating in weekly code reviews to uphold engineering standards.
- Built and owned CI/CD infrastructure using Jenkins and GitHub Actions that reduced deployment time from 2 hours to 15 minutes across 5+ environments, and integrated Datadog observability tooling to monitor production system health.

Projects

LLM Knowledge Retrieval and Q&A System | *Python, LangChain, OpenAI API, FAISS, PostgreSQL* — 2025
- Built a RAG system combining FAISS vector search with OpenAI embeddings for Q&A over large document corpora, instrumenting precision@k and MRR metrics to measure retrieval quality and iterating on indexing strategies until performance stabilized.

FitForm Analyzer | *Python, PyTorch, NumPy, Scikit-learn, Computer Vision* — Nov – Dec 2024
- Trained a CNN on 100K+ joint-position data points for posture classification, running ablations over architecture and loss functions to reach 88% accuracy, then applied post-training quantization to reduce inference latency by 40%.

MedEase: Healthcare Made Easy | *React, TypeScript, Python, TensorFlow, GCP, Firebase* — Aug 2023 – May 2024
- Designed and shipped a multilingual healthcare application end to end: React/TypeScript frontend, Python backend, Firebase database, and a TensorFlow intent classifier achieving 92% accuracy across 100+ languages using transfer learning and data augmentation.

Technical Skills

Languages: Python, Java, C/C++, Go, JavaScript/TypeScript, SQL
Backend & Systems: REST APIs, microservices, distributed systems, PostgreSQL, Redis, Kafka, Docker, Kubernetes, Terraform, Linux
ML & AI: PyTorch, TensorFlow, LangChain, OpenAI API, RAG, FAISS, Pandas, NumPy, Scikit-learn
Dev Tools: CI/CD (Jenkins, GitHub Actions), Datadog, JUnit, Git/GitHub, Cursor, Claude Code

Leadership & Activities

CMU ScottyLabs (CMU Maps) — Software Developer (Aug 2024–Present) | **Jane Street Discovery Day** | **NCWIT National Winner** (Apr 2024)
SkillsUSA National Cybersecurity — 1st Place, 3 consecutive years | Professional Speaker — Women in Tech & EdTech (50+ events) | CMU Quant Club